CHAPTER

4

Functional Occlusion

The basic objective of restorative dentistry is to develop the form, function and esthetics of the teeth to be in harmony with the stomatognathic system. These objectives can be achieved by understanding the nature of occlusion and functions of stomatognathic system.

The term 'occlusion' is defined as *"the static relationship between incising or masticating surfaces of maxillary and mandibular teeth"*.

Occlusion is also defined as *"the relationship between the occlusal surfaces of the maxillary and mandibular teeth when they are in contact"*.

The concepts of occlusion as emphasized by dental specialists are:

a. The concept of *balanced occlusion* followed by prosthodontists; the functional stability and effectiveness of complete dentures is maintained when there are simultaneous bilateral tooth contacts in lateral and protrusive excursions.

b. The concept of *morphologic occlusion* followed by orthodontists; the mesiobuccal cusp of maxillary first molar shall fall into the mesiobuccal groove of the mandibular first molar. Based on this maxilla–mandibular relationship, the occlusion is classified as:

- *Class I (neutro-occlusion or orthognathic):* The mesiobuccal cusp of the maxillary first molar falls into the buccal groove of mandibular first molar (Fig. 4.1) and one or more teeth are crowded, malposed, rotated, in crossbite or in open bite, etc.
- *Class II (disto-occlusion or retrognathic):* The mesiobuccal cusp of the maxillary first molar falls into the the embrasure between the mandibular second premolar and mandibular first molar (Fig. 4.2), i.e. the mandible has moved distally than in Class I occlusion.
- *Class III (mesio-occlusion or prognathism):* The mesiobuccal cusp of the maxillary first molar falls into the embrasure between the mandibular first molar and mandibular second molar, i.e. the mandible has moved mesially than in Class I occlusion (Fig. 4.3).

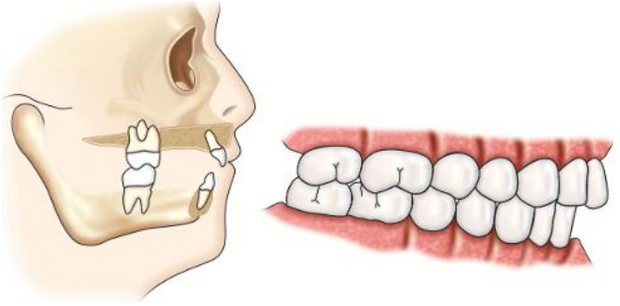


Fig. 4.1: Class I occlusion or normal occlusion

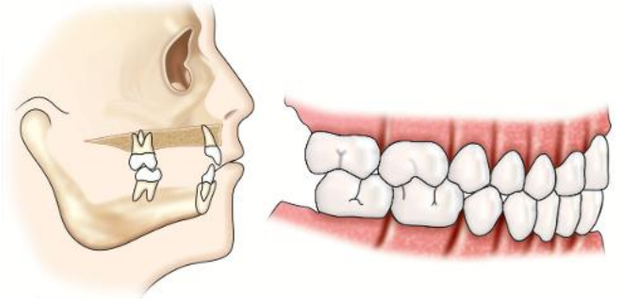


Fig. 4.2: Class II occlusion or disto-occlusion

c. The *concept of functional/physiologic occlusion*, followed mainly in restorative dentistry and periodontics; occlusion, which can function efficiently without pain and remains in a state of health

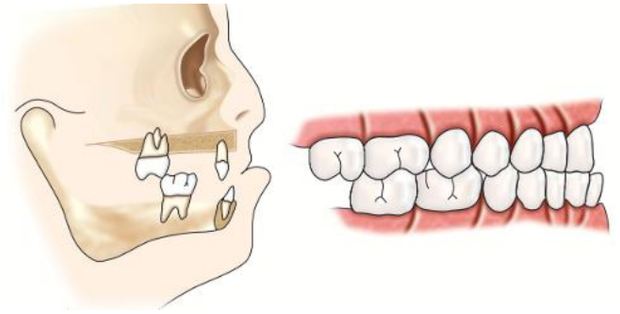


Fig. 4.3: Class III occlusion or mesio-occlusion

regardless of the relationship between the maxillary and mandibular teeth.

The characteristic features of functional occlusion are:

- Centric occlusion should produce stable maxillo-mandibular relationship at correct vertical dimension. There should be an acceptable inter-occlusal distance between the vertical dimension at rest and during occlusion.
- Bilateral simultaneous contact between the supporting cusps and respective fossae/marginal ridges to ensure even and axial distribution of occlusal forces. In centric occlusion, the condyles should be symmetrically placed in their fossae. *(Structural harmony between occlusion and the temporomandibular joints helps in achieving neuromuscular harmony).*
- Initial contact in centric relation should preferably be bilateral; the subsequent 'slide in centric' should be within 1.0 mm in an anterior direction.
- There should be harmony between the cuspal inclines during working and non-working movements of the condyles.
- No premature contact should be present in any mandibular movements.
- On lateral excursion, either canine guidance or group function should be present on working side, with no contacts on non-working side.
- On protrusion, there should be posterior dis-occlusion (no posterior contact).
- The masticatory forces should be distributed over as many teeth as possible in the intercuspal position (ICP).
- Bilateral contacts should be present between the posterior teeth in the retruded cuspal position (RCP).

TERMINOLOGY

Ideal occlusion: The most dynamic position, arrangement and relationship of one tooth with another tooth, of one arch with another arch and of both the arches with the base of the skull so as to perform the optimal functions of mastication, phonetics and esthetics maintaining the integrity and longevity of the individual tooth and the stomatognathic system.

Normal occlusion: The arrangement of the natural teeth, may be deviant, can be normal for that individual, as long as the function is satisfactory (no pain/discomfort and without damage to supporting structures).

Functional occlusion: The contacts of maxillary and mandibular teeth during mastication and deglutition.

i. *Lateral functional occlusion* (mandible moves laterally) is predominantly guided by canines (may involve sharing of contract by other posterior teeth in the functional working segment).
ii. *Protrusive functional occlusion* (mandible moves forward), all mandibular anterior teeth will contact along the palatal inclines of maxillary anterior teeth.

Non-functional occlusion (balancing contacts): A balanced non-functional occlusion may be necessary to provide adequate tooth guidance for mandibular excursion.

i. *Lateral non-functional occlusion:* The contact occurs on the opposing inclines between the two zones of centric contact. The potential non-functional lateral contacts can occur on lingual inclines of mandibular facial cusps and facial inclines of maxillary palatal cusps.
ii. *Protrusive non-functional occlusion:* The contact may occur on either or both the right and left posterior segments.

The directions of movement of the mandibular buccal cusps, during working (W), non-working (NW), and protrusive (P) movements are shown in Figs 4.4 and 4.5, respectively.

Physiologic occlusion: The occlusion in harmony with the functions of masticatory system.

Non-physiologic occlusion: The occlusion which leads to or associated with traumatic lesions or disturbances in supporting structures of teeth, muscles and temporomandibular joint.

Supporting cusps/centric holding cusps/stamp cusps are the palatal cusps of maxillary molars and bicuspids and the buccal cusps of mandibular molars and bicuspids. The incisal edges of the mandibular anterior teeth are also used in this designation. Centric

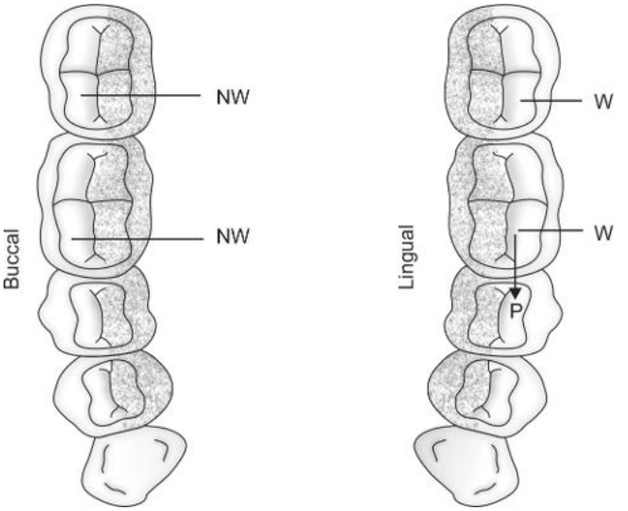


Fig. 4.4: The direction of movement of the mandibular buccal cusps (in arch) during working (W), non-working (NW) and protrusive (P) movements

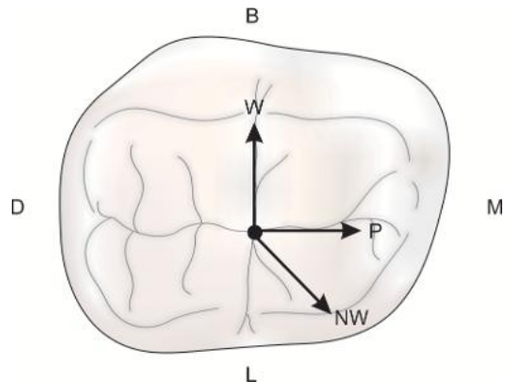


Fig. 4.5: Direction of movement of mandibular buccal cusps during working (W), non-working (NW) and protrusive (P) movements

stops are the areas of contact that the supporting cusp makes with opposing teeth, e.g. mesiopalatal cusp of maxillary first molar (supporting cusp) makes contact with central fossa (centric stop) of mandibular first molar.

Non-supporting cusps/non-centric holding cusps/shear cusps are located in the anteroposterior plane in facial and lingual embrasures or in the developmental grooves of opposing teeth creating an alternate arrangement when the teeth are in normal interdigitation.

Guiding inclines are the bucco-occlusal inclines (lingual inclines of buccal cusps) of maxillary posterior teeth, lingual inclines of maxillary anterior teeth and the linguo-occlusal inclines (buccal inclines of lingual cusps) of mandibular posterior teeth (guiding inclines determine the path of supporting cusps during normal and protrusive excursions).

Incisal guidance is the vertical and horizontal overlap of anterior teeth; causes the mandibular anterior teeth to track downward from the area of centric contact as they move toward the incisal edges of their maxillary opponents during protrusive and lateral excursions (Fig. 4.6A–C).

Condylar guidance is the path the horizontal rotational axis of the condyles travels during normal mandibular opening. The components of condylar guidance are:

i. *Contour of articular eminence of the temporal bones:* Articular eminence is the surface of temporal bone that slopes downward and forward from glenoid fossa of temporomandibular joint. The condyles move downward along the eminence as the mandible moves forward. This forward movement is called 'translation'. Both condyles translate simultaneously along their eminences in protrusive functional movement.

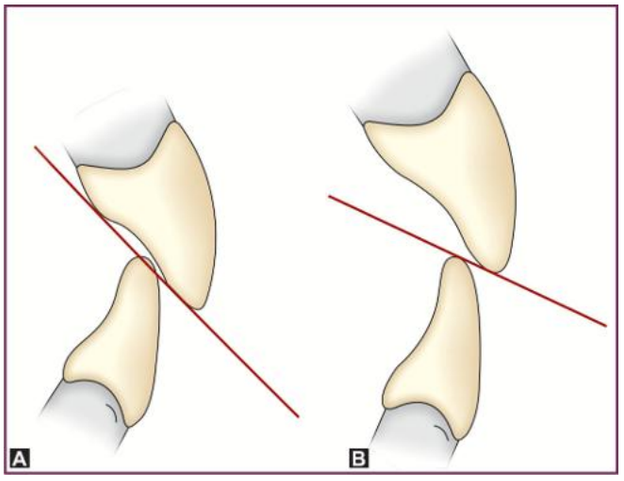


Fig. 4.6: Incisal guidance: **(A)** Steep; **(B)** Shallow



Fig. 4.6C: Canine and incisor guidance (working tooth guidance involving canines, central and lateral incisors)

ii. *Bennett shift/Bennett movement:* It is the lateral bodily shift of mandible towards the working side in function. The amount and the timing of lateral shift influence the pattern of tooth contact during lateral excursion (Fig. 4.7).

iii. *Intercondylar distance:* The intercondylar distance affects the path of lateral functional movement of mandible since it determines the location of vertical axis of rotation in relation to mandibular arch (Fig. 4.8).

Cusp angle is the angle made by the slopes of the cusp with a plane that passes through the tips of the cusps and is perpendicular to the line bisecting the cusp.

Plane of occlusion is an imaginary surface that touches the incisal edges of incisors and tips of occluding surfaces of posterior teeth. This surface is not flat but curved.

Curve of Spee is an anteroposterior curvature of occlusal plane of the posterior teeth, when a line is

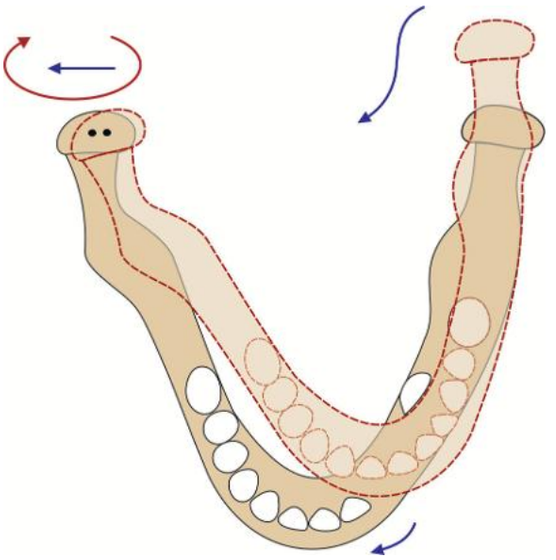


Fig. 4.7: Bennett movement and Bennett angle

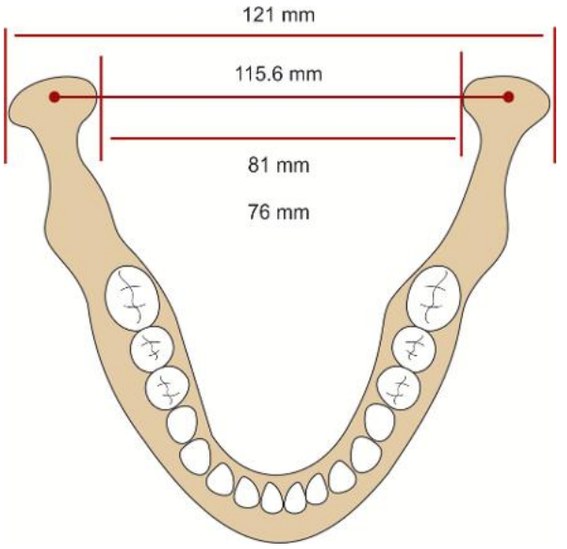


Fig. 4.8: Intercondylar distance

drawn from the tip of the mandibular canine along the tips of buccal cusps of mandibular posterior teeth (Fig. 4.9).

Curve of Wilson is the transverse curvature of occlusal plane of posterior teeth, when a line is drawn from cusps tips of one side to the cusps tips of the other side (Fig. 4.10).

CENTRIC RELATION

The centric relation is defined as *"maxillomandibular relationship in which the condyles articulate with the thinnest avascular portion of their respective discs with the complex in an anterior superior position against the slopes of the articular eminences. This position is independent of tooth contact and clinically discernible when the mandible is directed superiorly and anteriorly. It is restricted to a purely rotatory movement about a transverse horizontal axis"*.

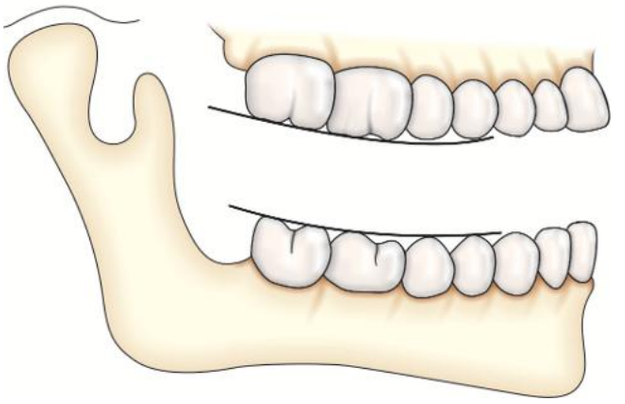


Fig. 4.9: Curve of Spee

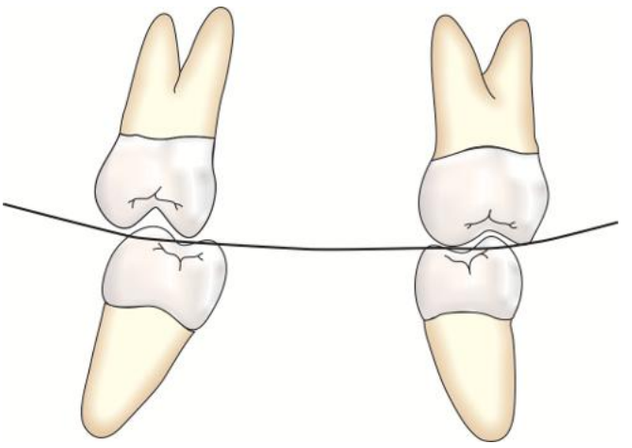


Fig. 4.10: Curve of Wilson

The normal relationship of mandibular condyle to glenoid fossa and related structures is shown in Fig. 4.11.

It is believed to be a precise location; whereas, a few authors believe in some freedom from this position [freedom in the anteroposterior direction is referred to as 'long centric'; and in the mediolateral position, referred to as 'wide centric' (Fig. 4.12)].

In centric relation, the mandible is in its terminal position and will rotate along a horizontal axis, which is referred to as the *'terminal hinge axis'*. The condyles rotate around terminal hinge axis and the lower incisors also arc about 20–25 mm. This arc of movement is termed as the *'terminal arc of closure'*. The teeth make

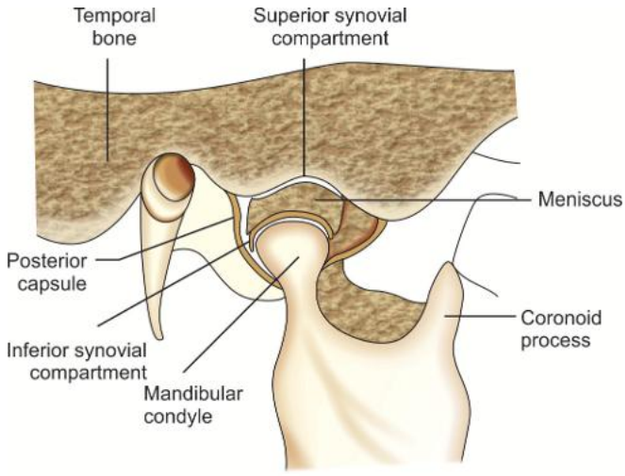


Fig. 4.11: Anatomy of temporomandibular joint (schematic representation)

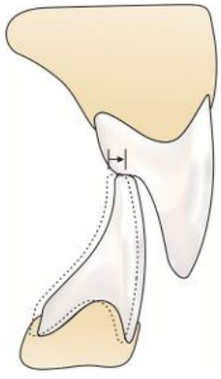


Fig. 4.12: Centric relation and the possible freedom of movement

an initial contact during this closure, called the *'retruded contact position'*.

CENTRIC OCCLUSION AND MAXIMAL INTERCUSPAL POSITION (MIP)

Centric occlusion is defined as *"the occlusion of opposing teeth when the mandible is in centric relation (maximum intercuspation of teeth which is in harmony with the neuromuscular mechanism)"*.

Maximal intercuspal position is defined as *"the complete intercuspation of opposing teeth independent of condylar position, sometimes referred to as the best fit of the teeth regardless of the condylar position"*; also called maximal intercuspation.

The maximal intercuspal position (MIP) does not coincide with centric relation/retruded contact position in about 90% of the individuals; in 10% of the individuals, the maximal intercuspal position (MIP) coincides with centric relation/retruded cuspal position.

In centric occlusion or maximal intercuspal position, the maxillary teeth overlap the mandibular teeth. The horizontal overlap of maxillary incisors to mandiular incisors is called *overjet* and the vertical overlap of maxillary incisors is known as *overbite*. The horizontal overlap of the posterior teeth is called buccal overjet and will help in preventing the cheek or tongue bite. The normal occlusal anatomy of maxillary first molar and mandibular first molar is depicted in Figs 4.13 and 4.14, respectively.

The cusps of posterior teeth are divided into four inclines. Each incline is named by the direction it faces. In mandibular molars, the inclines will be: Mesiobuccal incline, distobuccal incline, mesiolingual incline and distolingual incline. For convenience, the inclines are further divided into two: Outer incline and the inner incline. The inner inclines are surfaces between the cusp tip and the central fossa. Outer inclines are the surfaces that face outward towards the cheeks or tongue starting from the cusp tip.

Contacts that occur on cuspal inclines are called *poded contacts*. Stable centric contacts may contact two, three or four opposing inclines and hence are termed bipoded contacts, tripoded contacts and quadrapoded contacts, respectively.

Mostly there is an anterior and upward slide of the mandible from centric relation to centric occlusion. This slide, which occurs on the inclines of the premolars

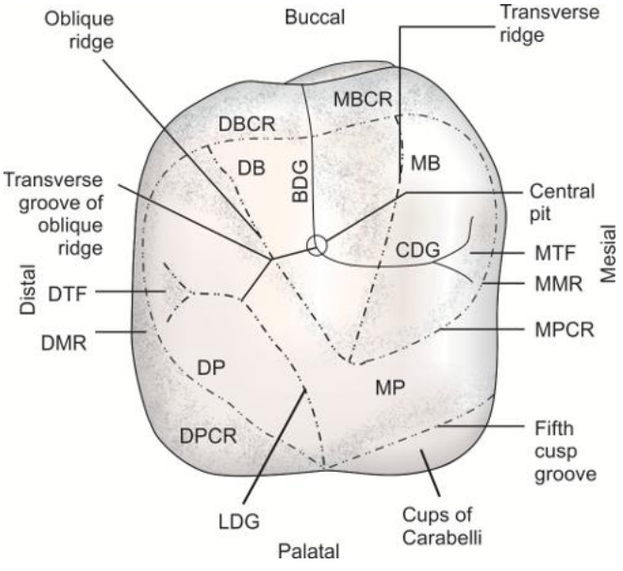


Fig. 4.13: Occlusal anatomy of maxillary 1st molar: MB: Mesiobuccal cusp, DB: Distobuccal cusp, DP: Distopalatal cusp, MP: Mesiopalatal cusp, BDG: Buccal development groove, LDG: Lingual developmental groove, CDG: Central developmental groove, MTF: Mesial triangular fossa, DTF: Distal triangular fossa, MBCR: Mesiobuccal cusp ridge, DBCR: Distobuccal cusp ridge, DPCR: Distopalatal cusp ridge, MPCR: Mesiopalatal cusp ridge

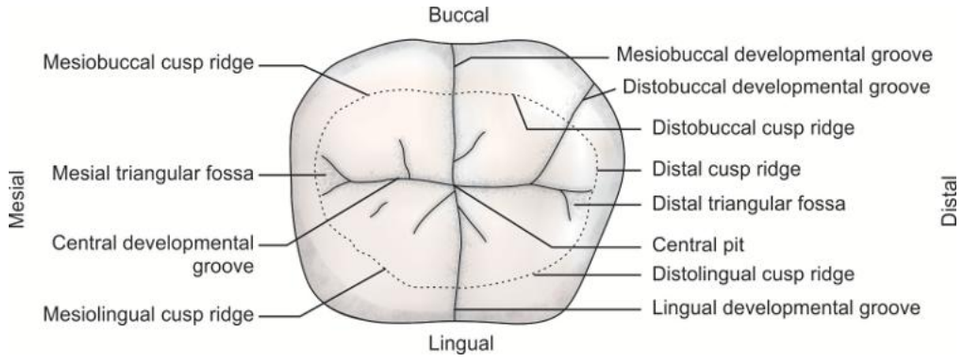


Fig. 4.14: Occlusal anatomy of mandibular first molar

and molars that come in contact during motion, seldom increases 1 mm. As the mandible slides from centric relation to centric occlusion, the condyles move downward and forward. In rare cases, the centric occlusion coincides with centric relation and there is no slide of the mandible. If the patient and the musculature is relaxed, the clinician can guide the mandible into centric relation.

Contact Areas: Occlusal Surface

In an ideal arrangement of teeth, the facial to lingual zones of potential contact of maxillary posterior teeth are (Fig. 4.15):

Zone 1: Lingual inclines of facial cusps—lateral functional contact (facial range)
Zone 2: Central groove area—centric contact (facial range)

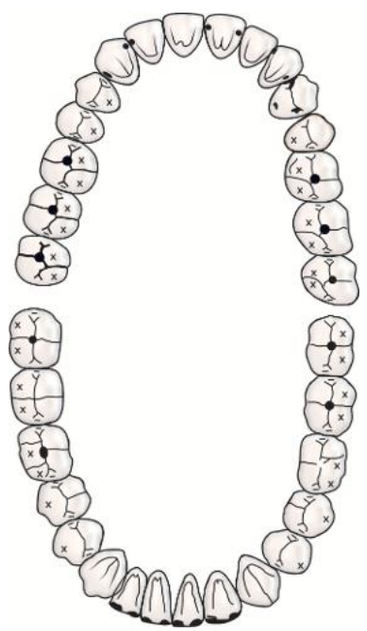


Fig. 4.15: Contact areas of mandibular and maxillary teeth in centric (x), contacts (•) on the opposing teeth

Zone 3: Facial inclines of palatal cusps—lateral non-functional contact
Zone 4: Lingual cusp tips—centric contact (lingual range)
Zone 5: Lingual inclines of palatal cusps—lateral functional contact (lingual range)

In an ideal arrangement of teeth, the facial to lingual zones of potential contact of mandibular posterior teeth are (Fig. 4.15):

Zone 1: Facial incline of facial cusp—lateral functional contact (facial range)
Zone 2: Facial cusp tips—centric contact (facial range)
Zone 3: Lingual inclines of facial cusps—lateral non-functional contact
Zone 4: Central groove area—centric contact (lingual range)
Zone 5: Facial inclines of lingual cusps—lateral functional contact (lingual range)

The cusp tips and lingual inclines of lingual cusps of mandibular posterior teeth normally have no contact potential. Similarly, the cusp tip and buccal inclines of buccal cusps of maxillary teeth normally have no contact potential.

OCCLUSAL DESIGNS/SCHEMES

The basic designs/schemes of occlusion are:

a. Balanced Occlusion

The balanced occlusion is a prosthodontic concept, which dictates that a maximum number of teeth should contact in all excursion positions of the mandible. This is particularly useful in complete denture construction, in which contact on the non-working side is important to prevent tipping of the denture. The concept was applied to natural teeth in complete occlusal rehabilitation. An attempt was made to reduce the load on individual teeth by sharing the stress among as many teeth as possible. As a result of the multiple tooth contacts, there was excessive frictional wear on the teeth.



Fig. 4.16: Canine guidance (working tooth guidance involving canines)

b. Mutually Protected (Canine Protected) Occlusion

The posterior teeth prevent excessive contact of the anterior teeth in maximum intercuspation, and the anterior teeth disengage the posterior teeth in all mandibular excursive movements. The theory of canine protected occlusion is based on the impression that the canine tooth is the most appropriate tooth, which guides the mandibular excursion (Fig. 4.16).

- In intercuspal position, primary occlusal load is absorbed by posterior teeth (Fig. 4.17A to C). Anterior teeth contact lightly and do not bear the load.
- During protrusion, the overjet and overbite should cause disocclusion of the posterior teeth (Fig. 4.18A, B).
- During lateral excursion, the overjet and overbite relationship of the contacting teeth on working side should cause disocclusion of teeth on non-working side (Fig. 4.19A to C).
- On the working side, either the canines (canine protected) or the canines, premolars and molars may be in contact (group function).

c. Group Function Occlusion

Group function occlusion (unilateral balanced occlusion) is a widely accepted method of tooth arrangement in restorative dental procedures (Fig. 4.20).

This occlusion calls for all teeth on the working side to be in contact during lateral excursion. On the other hand, the teeth on the non-working side are contoured to be free of any contact. The group function of the teeth on the working side distributes the occlusal load.

OCCLUSION AND RESTORATIVE DENTISTRY

Analysis of Occlusion Prior to Restoration

It is imperative to examine and analyse the pre-existing occlusion prior to any restoration. The occlusion can be analyzed by clinical examination and analysis of occlusal functions. The disturbances in occlusion should be eliminated/corrected prior to the restoration of the teeth.

The healthy conversation is important to receive accurate information from the patient regarding occlusal disharmony. Certain questions, which can lead to initial diagnosis, are:

- Any difficulty in chewing.
- Clicking of teeth or joints during chewing.
- Any difficulty in opening the mouth.
- Any parafunctional habits, such as, nail biting, clenching, grinding, etc.
- Bleeding gums while chewing.
- Discomfort with the previous restorations, if any.

Clinical Examination

The clinical examination should observe:

- Pattern of occlusal wear.
- Malposition of individual tooth, like tipping of teeth.
- Interferences, if any, in the cusp–fossa or cusp–ridge relationship.
- Presence of plunger cusps and/or open contacts.
- Tilting, supra-eruption or malposition of teeth.
- Periodontal conditions of teeth to include—tenderness, hypermobility or pathological migration.

The clinical information is further strengthened by radiographic examination and the study of diagnostic casts.

The intraoral radiographs can depict the changes in the alveolar bone structure and also reveal the extent of bone loss. The study casts help us in gathering information about the morphology of teeth and intra-/



Fig. 4.17: Intercuspal positions



Fig. 4.18: **(A)** Protrusive movement; **(B)** Disocclusion of posterior teeth in protrusive movement



Fig. 4.19: **(A)** Left lateral movement; **(B)** Working side contacts; **(C)** Disocclusion on non-working side

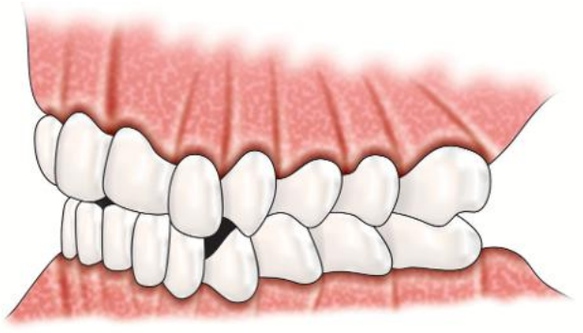


Fig. 4.20: Group function guidance (working tooth guidance involving canines, premolars and molars)

inter-arch relationship. They also help us in observing the presence of wear facets, tipping of teeth, supra-eruption, migration, open contacts and the presence of plunger cusps.

Articular Analysis

The occlusal contacts in maximal intercuspal position and during functional/non-functional movements are difficult to be examined intraorally. Casts mounted on articulator, which simulates mandibular movements, help in examining the occlusion from all aspects.

The types of articulators and the process of articulation are briefly described.

An articulator is a mechanical device, which represents the temporomandibular joints and the jaw members to which maxillary and mandibular casts may be attached to simulate jaw movements.

An articulator has an upper member and a lower member to which the maxillary and the mandibular casts are attached using dental plaster.

Articulation is defined as *"the static and dynamic contact relationship between the occlusal surfaces of the teeth during function"*.

Articulators are broadly classified as Arcon and Non-Arcon articulators, based on the attachment of the condylar elements to either the lower or upper member of the articulator. When the condylar elements are attached to the lower member, the articulator is called as Arcon articulators and when it is attached to the upper member, it is called as Non-Arcon articulator.

Examples of Arcon articulator: Hanau H2, Hanau wide vue, Panadent, etc.

Examples of Non-Arcon articulator: Hanau H, Trubyte, Dentatus, etc.

Types of Articulators

a. ***Hinge articulator:*** The hinge articulator is basically a cast holder. Only opening and closing movements are possible. There is no possibility of lateral or protrusive movements.

b. ***Plane line/mean value articulator:*** Mean value articulators permit a limited degree of protrusive and lateral movements used for fabricating individual crowns or a small bridge (Fig. 4.21).

c. ***Semiadjustable articulator:*** The semiadjustable articulator is routinely used in restorative dentistry (Fig. 4.22A, B). These articulators accept a facebow transfer to relate the maxillary cast to the hinge axis. Facebow correctly positions the maxillary cast spatially in three dimensions by way of the third reference point (orbital plane).

Advantages of semiadjustable articulator

- Orients the maxillary cast to the hinge axis of the articulator.
- Transfers the retruded path of closure or centric relation.
- Simulates the condylar guidance as it is in the patient.

d. ***Fully adjustable articulators:*** These articulators require pantographic tracing to customize the articulator for the individual requirements. These instruments are technique sensitive and are not used for restorative purposes.

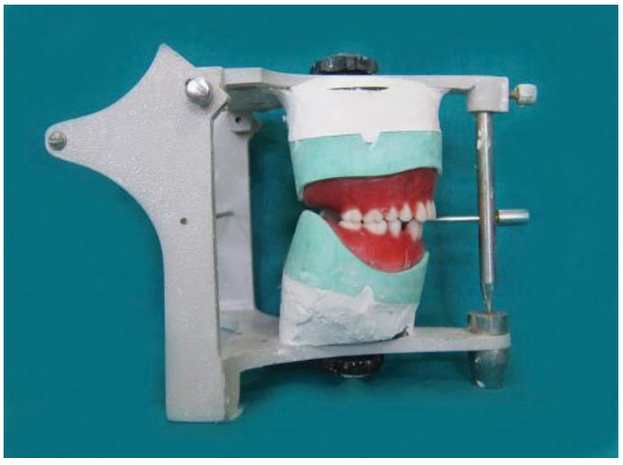


Fig. 4.21: Mean value articulator

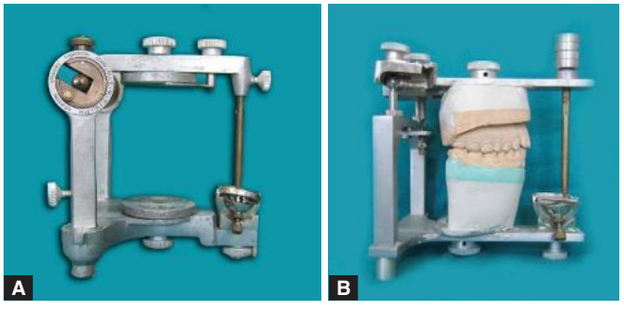


Fig. 4.22: Semiadjustable articulators: **(A)** Hanau; **(B)** Whipmix

Facebow Records

An instrument, used to transfer the relationship of maxilla and the terminal hinge axis from the patient to the articulator is called a facebow (Fig. 4.23A, B).

A facebow is a caliper-like device to record the spatial relationship of maxillary arch to some anatomic reference point and then transfer this relationship to an articulator; it orients the dental casts in the same relationship to the opening axis of the articulator.

Facebows can be broadly grouped as arbitrary facebows and kinematic facebows. Arbitrary facebows are used to transfer the orientation relation from the patient to the articulator. Kinematic facebow is used to record the true hinge axis.

Arbitrary location of the hinge axis is adequate for transferring the orientation relation for most of the restorative procedures. The arbitrary hinge axis is located 13 mm anterior to the tragus on a line drawn from the center of tragus to the outer canthus of eye. This point is called Beyron's point (Fig. 4.24).

Facebows normally use orbitale or nasion as the third point of reference to orient the occlusal plane in its sagittal relationship to the horizontal plane (Fig. 4.25A, B).

Articulation of Casts

In the absence of any signs and symptoms of trauma from occlusion/temporomandibular joint disturbances, an interocclusal record made in maximal intercuspal position is used to mount the mandibular cast onto the articulator to develop the occlusal morphology of the restorations.

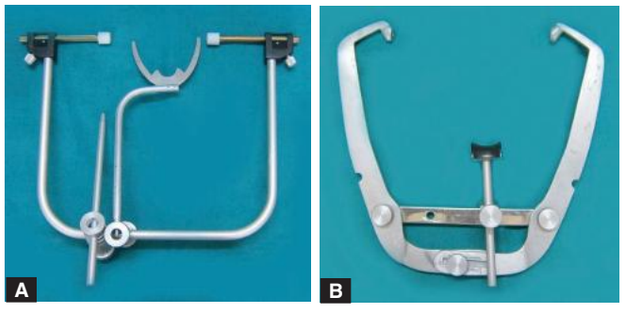


Fig. 4.23: Facebow: **(A)** Hanau; **(B)** Whipmix



Fig. 4.24: Beyron's point



Fig. 4.25: Facebow recording: **(A)** Hanau (orbital reference point); **(B)** Whipmix (nasion reference point)



Fig. 4.26: Cast along with interocclusal record mounted on articulator

Mounting the casts on a semiadjustable articulator using a facebow transfer and an interocclusal record are mandatory to refine the occlusion in lateral excursions and need to alter the vertical dimension (Fig. 4.26). When occlusion is to be developed in a static relationship, the maxillary and the mandibular casts can be mounted on a mean value articulator.

Interocclusal Records

Interocclusal records are used to relate the mandible to the maxilla in a protrusive/lateral excursive position to customise the condylar guidances.

The interocclusal records should be made in a material, which is dimensionally stable and can relate the mandibular to the maxillary casts.

The ideal interocclusal record material should have the following properties:

- Soft and moldable.
- Should not displace the soft tissues or teeth.
- Should not guide the mandibular movements
- Dimensionally stable.

The interocclusal recording material is manipulated according to the manufacturer's instructions and is used to make the interocclusal record in the desired mandibular position (Fig. 4.27).

Occlusal Records for Restoring a Single Posterior Tooth

The stepwise procedure is as follows:

- Two sheets thickness of baseplate wax is cut to cover the width and length of the prepared tooth.
- Cut wax is softened in warm water.



Fig. 4.27: Adjusting interocclusal records during mandibular movements

- The softened wax is placed over the prepared tooth; mandible is guided into the intercuspal position.
- The wax is kept at mouth temperature for one minute.
- Cool with air-water spray and remove it from the oral cavity.
- Examine the interocclusal record for any defect; can be repeated, if need be (Fig. 4.28 A to C).

Occlusal Records for Restoring all Posterior Teeth in a Single Arch

The stepwise procedure is as follows:

- Establish the vertical dimension of occlusion before the tooth preparation.
- An interocclusal record can be made with two sheets thickness of baseplate wax or any other interocclusal recording medium.
- Soften and place the wax over the prepared teeth.
- Guide the mandible into centric relation.
- The wax is kept at mouth temperature for one minute.
- Cool with air-water spray and remove it from the mouth.
- Examine the interocclusal record for any discrepancy; can be repeated, if need be (Fig. 4.28 A–C).

PRINCIPLES OF EDEC

A few authors opine following EDEC (examine, design, execute and check the occlusion) principle for producing the desired restoration. The principle is useful in both direct and indirect restorations.

EDEC Principle for Indirect Restorations

In case of indirect restoration, the laboratory technician is also involved in decision making. The operator records the occlusion and the same is transferred accurately to the technician. During the time gap of fabrication of the restoration, the operator has the responsibility of maintaining the occlusion with provisional restorations.

Examine

The occlusion is examined prior to the start of preparation. Both static and dynamic occlusions are to be analyzed, using appropriate articulating papers and/or foils. The cast models with marked occlusal problems are sent to technician for analysis. The anatomical information can be transferred by using photographs, bite records or by occlusal sketching. Both static and dynamic records are taken and transferred to the technician.

Design

The designing of tooth preparation depends upon the material to be used for fabrication of the restoration. Minor adjustments in the model can be carried out by the technician after due discussion with the operator. Such adjustments in the model is named as 'model grooming'. Many authors, however, do not favor this type of adjustments. They are of the view



Fig. 4.28: Interocclusal records: **(A)** With silicone; **(B)** With wax; **(C)** With wax (clinical)

that any adjustment in the model will lead to defective restoration; preferably the preparation is to be modified accordingly and the impression is repeated.

Execute

The preservation of patient's occlusion is mandatory, till the time the permanent restoration is placed. Interim restorations provide the necessary relief, duplicating the occlusion and maintaining till the procedure lasts.

Check

The occlusion is to be checked thoroughly during the preparation and also after placing the restoration. The occlusal interferences, if any, are to be noted and cleared. The final finished and polished restoration should simulate the functional occlusion of the patient.

RESTORING INDIVIDUAL TOOTH

The clinical features of importance are:

- All the restorations should be in harmony with the occlusion and should contribute for the health of the stomatognathic system.
- The restoration should not introduce any premature contacts; the embrasures should be optimally placed, so as to prevent caries and periodontal problems.
- The outline of the procedure should be planned to retain the natural crown contours and guide the development of occlusal morphology of the restorations.
- The intact buccal and lingual surfaces of the crown guide the accurate contouring of the axial surface of the restoration and help preserve the gingival health.
- Complete coverage restoration and onlays should be preferred only when the occlusal morphology needs to be re-established as in grossly destructed teeth.
- Knowledge of normal occlusal contacts of maxillary and mandibular teeth before beginning the tooth preparation is essential to reproduce the normal contacts or to modify the unfavorable contacts.
- Before beginning the tooth preparation, examine the occlusal contacts and determine, if they should be included or excluded in the outline form. Occlusal contacts should not fall on the restoration-tooth interface.
- The principle of establishing tripod contacts can be easily developed on cast restorations fabricated by indirect technique. Tripod contacts are desirable to provide retention and stability of the restoration (Fig. 4.29). However, carving a tripod contact intra-orally is difficult while placing a restoration. A few authors have advocated the establishment of a single point contact at the bottom of the fossa as an alternative to tripod contact (Fig. 4.30).
- The operator should confirm that the contacts are in harmony with the protrusive and lateral excursive movements (Fig. 4.31A, B).

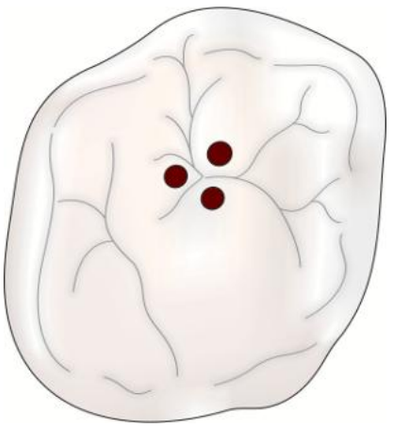


Fig. 4.29: Tripod contacts

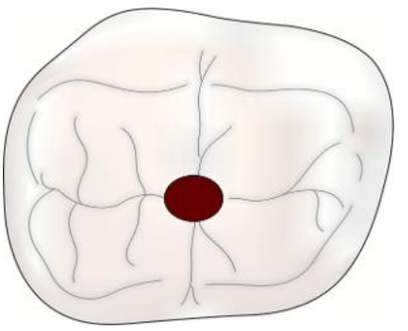


Fig. 4.30: Single point contact

Reasons for Restoring the Normal Anatomical Form of an Individual Tooth

- The correct relationship with adjacent teeth will give the best support against masticatory stresses, promote deflection of food through the embrasures and prevent food impaction.
- The correct relationship with opposing teeth during mastication will prevent deflective occlusal contacts, which might lead to pain, periodontal damage and fracture of teeth or restoration.
- The correct buccolingual contour will allow deflection of food over the free gingiva, thus protecting the periodontal tissues.
- Help in maintaining oral hygiene.
- Contribute for esthetics

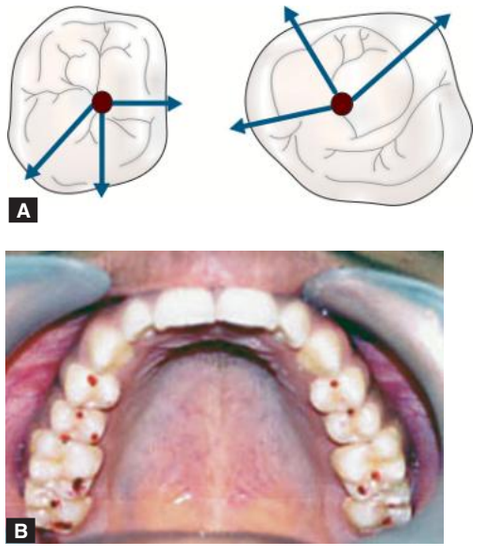


Fig. 4.31: **(A)** Restoration in harmony with excursive movements; **(B)** Visualization of occlusal contacts

Occlusal Consideration during Tooth Preparation

The intact buccal and lingual surfaces of the crown guide the accurate contouring of restoration and help preserve gingival environment. Hence, conservative preparation should be planned to avoid losing the guidance from natural contours of the teeth. Complete coverage restorations or onlays are indicated only when the cusps are severely damaged, fractured or when the occlusal morphology needs to be modified.

The biological width of the periodontium is normally about 2.0 mm from the crest of the alveolar bone. The health of the periodontium can be maintained by not encroaching on the biological width of the tooth.

During tooth preparation, an adequate amount of tooth material should be reduced to provide sufficient bulk of the restorative material of choice. The cervical areas should not be overcontoured. Particular care is to be exercised at the line angles, where the depth of the shoulder or chamfer is usually inadequate leading to overcontouring.

The overcontouring varies from individual-to-individual and also from one tooth to another in the same individual. The shape of the embrasures should be reproduced as in the functional occlusion. The facial and axial surface should be recontoured in the final restoration, only if the existing contour and embrasure do not protect food impaction.

To protect the functional cusp and improve the structural durability of the restoration, the outline form of the inlays and onlays should not be placed on the occlusobuccal line angle for the mandibular posterior teeth and occlusopalatal line angle for the maxillary posterior teeth. In such cases, the outline form can be extended to include the functional cusp in the preparation.

The depth of the occlusal preparation will vary depending upon the choice of the restorative material. For cast metal restorations, the functional cusps are reduced by about 1.5 mm and the non-functional cusps are reduced by about 1.0 mm. In case of metal ceramic restorations, the functional cusps are reduced by about 2.0 mm and the non-functional cusps are reduced by 1.5 mm.

To provide adequate retention for the complete veneer restorations, the occlusogingival length of the preparation should be at least 4.0 mm. In case of short crowns, the retention can be increased by placing grooves on the axial surfaces.

The adequacy of occlusal reduction can be ascertained by wax check bites.

Examining the provisional restoration under light will also help in determining the adequacy of the occlusal preparation.

Anterior Restorations

The horizontal and vertical overlap of the anterior teeth and the anterior guidance should be examined prior to restoration of the anterior teeth (Fig. 4.32).

The anterior guidance varies from one individual to another depending upon the type of occlusion and the jaw relation.

Canines play an important role in both canine protected and group function occlusion. Hence, care should be taken to retain the existing guidance; can be changed in rare cases as per need of the treatment plan.

Anterior guidance should provide smooth and even contacts on as many anterior teeth as possible. If a single tooth is to be restored, the existing guidance should be retained. When all the anterior teeth are to be restored, some changes in incisal guidance and position of teeth can be made, if necessary.

The relationship of mandibular anterior teeth to maxillary anterior teeth, especially in cases of Class II division 1 malocclusion, touches the dentogingival junction. In such cases, the mandibular incisors can be kept short to prevent trauma from occlusion.

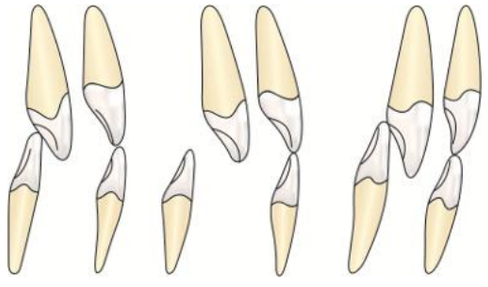


Fig. 4.32: Incisal guidance

The canine has a favorable crown/root ratio for absorbing occlusal forces. Its root configuration will also provide greater surface area and better proprioception than the adjacent teeth. The location of the canines is suitable for easily correcting the guidance, if necessary.

While restoring the canines into canine-guided occlusion, care must be taken to see that there is disocclusion of the posterior teeth during lateral excursions. While restoring the incisors, similar care should be taken to provide adequate space for the bulk of the material and to prevent the overcontouring of the teeth.

Before planning the posterior restorations, following features must be considered:

- Ascertain the stability of the intercuspal position.
- Determine the anterior tooth contact.
- Determine the adequacy of the existing vertical dimension of occlusion.
- Determine the evenness of the plane of occlusion
- Anticipated design of the tooth preparation.

Hindrances that may be caused in developing a harmonious occlusal scheme by the cusps of malpositioned teeth, supraerupted teeth or tilted tooth should be corrected before the tooth preparation.

Using an articulating paper, identify the cuspal contacts in the maximal intercuspal position and its movements into excursive movements. In case of intracoronal restorations, these contacts can either completely be included or excluded from the outline form of the restorations (Fig. 4.33). In case of extracoronal restorations, these contacts should be meticulously reproduced in the restoration.

The occlusal contacts located on the cuspal inclines or on the slopes of the ridges are undesirable because

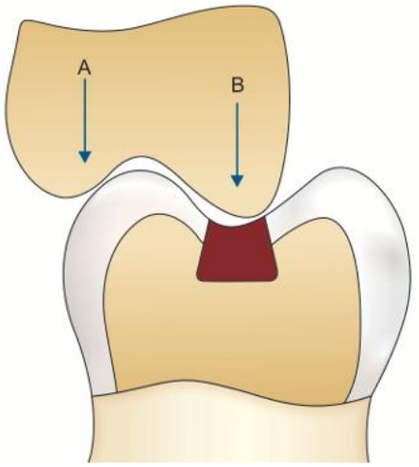


Fig. 4.33: Outline form **(A)** contact area excluded, **(B)** contact area included

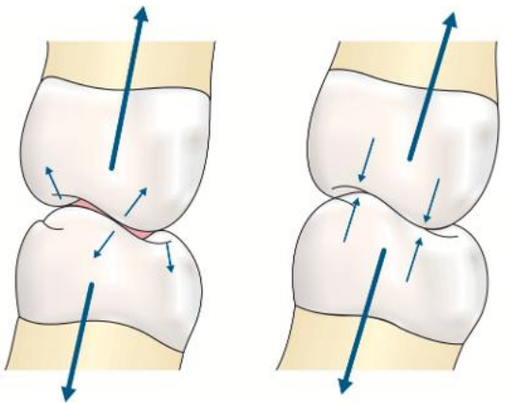


Fig. 4.34: Resultant forces

they create a deflective force on the tooth. Occlusal contacts should be placed such that the cusps should make a tripod contact with the opposing fossa or a point contact at the bottom of the fossa. Such contacts helps in directing the occlusal forces along the long axis of the tooth which is conducive to the health of the periodontal ligament (Fig. 4.34).

The choice of the posterior restorations depends upon the amount of remaining tooth structures and the clinical length of the crown. Indirect veneer restorations are preferred over direct restorations, whenever the direct restorations are large, has modifications, has an occlusal isthmus wider than half of the occlusal table or has repeatedly failed.

The restorations, may be single or multiple, must be in harmony with the planned occlusal scheme. An articulating paper/shimstock can be used to verify the occlusal contacts of the teeth to be restored and the adjacent teeth.

When a distal most tooth in an arch is to be prepared, it is important to accurately record the occlusal relation. Autopolymerizing acrylic resin copings can be used to verify the occlusal contacts of the prepared teeth intraorally and to verify the same on the articulated casts. The thickness of the coping can also be measured to verify the adequacy of the teeth preparation.

When the posterior teeth in any quadrant are lost, there is a tendency for the opposing teeth to supra-erupt, leading to uneven occlusal plane. The occlusal plane should be corrected prior to the placement of the restorations to prevent occlusal prematurities.

When restoring multiple units in a single arch, the occlusion can be developed in the maximal intercuspal position. Alternatively, when restoring all the teeth in a single arch, the occlusion can be developed in the retruded contact position. This approach can also be followed when there is a need to increase the vertical dimension, associated with short clinical crowns.

Interocclusal records are made in the retruded contact position, the casts are articulated on a semiadjustable articulator using a facebow transfer.

Provisional restorations are given to analyze the adaptation of patient to new occlusal scheme. Premature loss of one or a few provisional restorations indicates lack of occlusal harmony or poor retention.

Make the necessary corrections to eliminate the cause and place the provisional restorations again. When the patient is adapted to the new occlusal scheme, definitive restoration can then be fabricated to restore the occlusion.

Modifying the Occlusal Table

The occlusal table is the sum total of occlusal surfaces of all the posterior teeth; however, the occlusal table also includes areas beyond the palatal surfaces of upper teeth and the buccal surfaces of lower teeth.

In certain cases, this table needs to be modified or made narrower, which leads to:

- ***Reduction of force:*** When the occlusal table is made narrower, lesser force is applied during masticatory functions (Force is transmitted to all structures underlying the occlusal table, which include the restoration, the tooth structure and the periodontium.)
- ***Reduction of the effect of force:*** The direction in which the applied force is transmitted is governed by muscular activities and the area on which the force is applied. However, effect of the force may be reduced by narrowing the surface at which the force is applied, thus altering the direction of the resolved components.
- ***Reduction of torque:*** The tendency to rotate may be reduced by altering the point of application of the force relative to the fulcrum. The point of application of the force may be altered by modifying the occlusal table; indirectly depends upon design of the restoration.
- ***Facilitation of oral hygiene and reduction of soft tissue biting:*** Oral hygiene may be improved and cheek biting can be eliminated by altering the occlusal table.

Role of Contact Areas

Human teeth are designed in such a way that they contribute significantly to their own support as well as supporting the arch. Each tooth assumes the same responsibility in providing support that a brick assumes as a part of a wall. This arrangement relieves the supporting bone of much responsibility and the forces can be dissipated smoothly. A break in the continuity of the contact areas put additional pressure on the periodontal membrane and alveolar bone, which they are not designed to sustain. The teeth move into convenient positions, and eventually lead to breakdown of the supporting tissues. Interproximal contacts aid in shunting the food towards the buccal and lingual areas and thus prevent the food from pushing in between the teeth and thereby impinging on gingival tissues, which could predispose to periodontal problems and/or dental caries.

Dental restorations must reproduce the proper contact areas. The flat, open, improperly placed contact areas couple with rough or poorly polished surfaces will lead to failure of restoration. The opposing interproximal surfaces of restorations must be hard in order not to flow, flatten, wear or become abraded with use.

Hazards of Faulty Contacts

Faulty contacts are always detrimental to underlying periodontium.

The types of fault and their consequences (for details refer to Table 5.2, Chapter 5) are as follows:

a. Too broad contact, buccolingually or occlusogingivally, leads to:
   i. Change in tooth anatomy and the shape of interdental col. The normal saddle-shaped area is broadened. Since the col epithelium is nonkeratinized, it increases the area of susceptibility to periodontal disease.
   ii. Improper shunting of food in buccal and lingual directions because of narrow embrasures; hence, the chances of food impingement into the contact area is increased, resulting in inefficient mastication.

b. Too narrow contact, buccolingually or occlusogingivally, leads to:
   - Food impaction vertically and/or horizontally on col area;
   - Wide buccal and lingual embrasures in which normal oral hygiene measures may not gain access predisposing to greater food retention and plaque accumulation in embrasure areas.

c. Contact placed too occlusally will result in a flattened marginal ridge at the expense of occlusal embrasure.

d. Contact placed too buccally or lingually will result in a flattened restoration at the expense of buccal or lingual proximal wall.

e. Contact placed too gingivally will increase the depth of occlusal embrasure at the expense of the size of contact area or at the expense of broadening or impinging upon interdental col.

f. Loose contact creates continuity between embrasures and interdental col, leading to food impaction.

Role of Contours

The contours (convexities and concavities) of the tooth which should be reproduced in a restoration.

a. ***Facial and lingual convexities:*** Convex contours on the facial and lingual surfaces of the teeth afford protection and stimulation to the supporting structures during mastication. They direct the food towards the buccal vestibule, palate or tongue while stimulating the surrounding soft tissue by gentle massage. Overcontoured curvatures can create a favorable environment for the accumulation and growth of cariogenic and plaque bacteria at gingival margin, apical to the height of contour; subsequently, results in chronic inflammation of gingiva.

b. ***Facial and lingual concavities:*** Concavities occlusal to the height of contour, whether present on anterior or posterior teeth, are involved in occlusal, static and dynamic relations; they determine the pathways for mandibular teeth into and out of centric occlusion. Deficient or dislocated concavities can lead to premature contacts during mandibular movements, which could inhibit the physiologic capabilities of these movements. Excessive concavities can invite extrusion, rotation or tilting of occluding elements into non-physiologic relation with opposing teeth.

c. ***Proximal contours adjacent to the contact area:*** Proximal curvatures adjacent to the contact area form 'V' shaped spillway spaces, called embrasures. Embrasures serve as pathways for the passage of food, which is returned to the occlusal surfaces by the movement of the facial tissues and the tongue. Food is prevented from accumulating on the occlusal surface. Occlusal embrasures prevent food impaction into the contact area. All occlusal, buccal, lingual and gingival embrasures should be restored. Too narrow embrasures predispose teeth and supporting structures to heavier stresses. Too wide embrasures offer little protection to the underlying soft tissue.

The readers are advised to consult Chapter 5 for details on contacts and contours.

Role of Marginal Ridges

The marginal ridges play an important role in withstanding and dissipating the occlusal stresses. The correct form of marginal ridge compatible with the adjacent tooth and also with its own surroundings is important during carving of posterior restoration. The absence of marginal ridge, or marginal ridge with improper height can lead to altered dissipation of forces; subsequently, damaging the underlying periodontium. The details as how these forces act on marginal ridges and how the altered marginal ridges deviate the forces are given in Chapter 12.

Tooth Wear and Restorative Materials

Occlusal forces lead to wear of enamel. The wear is slow, if occlusal forces are appropriately transmitted to underlying bony tissues. The pattern of wear varies depending upon various factors. Non-uniform wear of opposing teeth is quite common, when one teeth is restored with a restorative material whose wear resistance is different as compared to that of enamel. Rarely, the wear resistance of a restorative material equals the wear resistance of enamel.

At present, no restorative material is available, which wears at the same rate as enamel.

Hypothetically, if two restorative materials, which wear at a slower rate than the natural teeth are placed so as to oppose each other in a dentition, the restorations will produce occlusal interferences at a later stage.

Non-wearing materials opposing each other can lead to natural teeth wear during contact in lateral and protrusive movements. Conversely, if the materials wear faster than the teeth, the opposing cusp might over-erupt into the worn material. In lateral excursion, this cusp might then come in contact with an opposing cusp; and if weakened by caries, etc., may lead to fracture.

Murphy investigated the compensatory mechanisms associated with advanced tooth wear. He reported that 50% compensation resulted from continuous eruption (in the form of cementum apposition with root lengthening and differential alveolar bone deposition with reduction in socket depth). A 10% compensation resulted from general bone growth and 40% uncompensated as loss of occlusal face height. He postulated that the compensatory mechanisms occurred independently to those of wear, so that occlusal face height remained constant of wear.

Occlusal interferences can develop through differential wear patterns and unmatched compensatory mechanisms.

Since wear defects are not repaired automatically, the operator should replace and maintain the configuration of teeth in accordance with the functional activities.

RECORDING OCCLUSAL RELATIONS FOR INDIRECT RESTORATIONS

The occlusal relations are to be recorded so as to reproduce the same on the pattern and the casting for fabrication of indirect restoration. The occlusal relations during various movements of mandible are

simulated on the articulators and are recorded. It has been demonstrated that individuals close their jaws from the rest position to the intercuspal position (ICP) along a habitual path of closure. An altered ICP and mandibular path of closure can be caused by deflecting contacts resulting from restorations, tooth loss, tooth movements or differential wear.

Recording Mandibular Movements

The stages of recording mandibular movements are:

i. The first stage for any occlusal recording is to place the mandible in centric relation.
   The following points are important (verify centric relation):
   - The condyle is in the superior most position within the fossa. This can be achieved by physically guiding the mandible upwards, while manipulating it backwards from its habitual location. This act should continue until no further upward movement of mandible occurs.
   - The mandible in centric relation position moves in a rotating arc. This can be verified by rotating the mandible up and down in very short arcs.
   - Muscles should be in complete harmony with each other.
   - There should not be any tooth contact at this position.

ii. The second stage is to locate centric occlusion. The patient is asked to bite, while the operator guides and supports the horizontal position of mandible.

iii. The third stage defines lateral excursions, which start from centric occlusion. Supporting pressure is applied at the angle of the mandible on the working side to keep the condyle seated within its fossa.
   The mandible should be guided until the cuspid on the working side is edge-to-edge and ready to disocclude. The action is carried out twice, once for each side. The application is repeated for non-working side.

iv. The fourth stage is the protrusive movement, which starts from centric occlusion and the patient is asked to guide the teeth forward. There should be superoanterior guiding with light supporting pressure at the angle of mandible while the patient is protruding it.

Transferable Records

These records are used for fabricating restorations outside the oral cavity. They are mainly fashioned in two steps. First step is making the record intraorally and the second step involves transferring the information to the articulator. The articulator movements create the same effect on the restoration as the movements of the mandible.

Both static and the dynamic relations are to be recorded and transferred to the articulator.

A. *Static Relationship Transferable Records*

These records capture the relationship of the maxillary and mandibular teeth at border locations of the mandibular movement path. They are only transferable to semiadjustable articulator, which is capable of moving from one border location to another with a standard path in between.

These records are also used to correlate upper and lower casts at centric relation or centric occlusion in any type of articulator (hinge, semiadjustable or fully adjustable).

a. ***Use of hinge articulator:*** This is indicated when the occlusion is stable and with adequate holding cuspal elements on both the sides. The only record needed for hinge articulator is the facial centric occlusion record. The opposing cast is occluded to the mounted cast via facial record and the cast then attached to the articulator.

b. ***Use of semiadjustable articulator:*** Here a facebow record is needed. The record capture the relationship of maxilla to the cranium. For these, three reference points are necessary—one in each direction at the cranial part of stomatognathic system. Joining these three points creates a plane close to hinge axis.

A wax carried in special bite-fork is indented with maxillary teeth at three or four reference points and the impression represents the maxilla. Creating the assembly having these four point contacts will formulate interfacebow record. This assembly is attached to the articulator. The facebow is removed and the upper cast is correlated with lower cast using following steps:

- A centric relation record is used to join the upper and lower casts.
- Held at this relationship, the lower cast is attached to the lower arm of the articulator with a non-expanding gypsum.
- After setting of gypsum, left condylar path controlling screws are loosened.
- The right excursive records are made between the casts and the left condylar path and also for right condylar path.
- Next protrusive excursive records are made.
- Holding the relationship, the incisal guidance is adjusted according to cast's anterior teeth, overjet, overbite and lateral disclusion capability.

B. *Dynamic Relationship Transferable Records*

The dynamic relationship records are created by functionally generated path, especially when single

or a few teeth in an arch are to be restored. The other methods, viz. pantographs and stereographic tracing are useful where vertical dimensions are to be modified (full mouth or large number of posterior teeth).

Functionally generated path (FGP) interocclusal record is useful when checking occlusal clearances, especially in areas difficult to visualize, such as in the central groove or the lingual cusp regions.

To make an FGP interocclusal record, first dry the preparation (free of viable moisture). Then lightly press a portion of softened, low fusing inlay wax over the prepared tooth (teeth) and immediately request the patient to close and slide the teeth in all directions.

During the mandibular movements, observe to verify that:

- The patient moves in right lateral, left lateral and protrusive movements.
- The adjacent unprepared teeth are in contact with opposing teeth.
- The wax in the cavity is stable.
- Carefully remove the wax. Hold it across the light and note the degree of light transmitted.

The light is a good indicator of thickness of wax. Insufficient thickness calls for more reduction in the indicated area before proceeding.

Such records provide information how to form occlusal surfaces and position occlusal contacts on the restoration; however, how these contacts might function during mandibular movements is not clear.

If case records of mandibular movements are required, then full arch cast is prepared and transformed to semiadjustable articulator. The FGP interocclusal record works well when one or two teeth are prepared; whereas, full arch casts are preferred when more than two prepared teeth are involved. FGP interocclusal records can register the interactions of anterior guidance, horizontal condylar guidance, cusp inclines and occlusal contact positions.

The use of full-arch casts mounted on a semi-adjustable articulator is recommended when restoring a large portion of the patient posterior occlusion with cast metal restorations.

Non-Transferable Records

Such records are used only for diagnostic or verifying purposes.

The following features should be noted before a final restoration.

- Uneven, non-symmetrical attrition of occluding teeth.
- Supraeruption, tilting, rotation or bodily movements of a tooth.
- Undercontoured (overcarved) occluding restoration leading to occlusal and possibly lateral displacement of opposing cuspal elements.
- Periodontal diseases facilitating tooth movement, especially laterally.
- Insufficiently restraining cusp-fossa or cusp-marginal ridge relationship allowing repositioning of teeth.
- Plunger cusps against marginal ridges separating them or the same cusp against one side of tooth tilting it laterally.

Checking the Casting for Occlusion

When the casting is satisfactorily seated on the tooth, ask the patient to close in maximum intercuspal position (ICP) and inspect the adjacent teeth.

After drying the teeth, insert a strip of articulating paper and request the patient to close and tap the teeth in intercuspal position. Remove the paper, and examine in the light for evidence of any areas of penetration. The heavy markings on the castings (metal-colored spots in the center of the marks) should be noted. Such heavy contacts should be reduced with suitable abrasive stones while carefully observing the following fundamental concepts for equilibration of occlusion.

- The space observed between opposing wear facets of adjacent unprepared teeth is an indication of the maximum amount of vertical reduction of the casting required.
- Often the 'high' occlusal contacts are too broad and extend onto cusp or ridge slopes. In such cases, grind away the most incorrect portion of the incline contact (a deflective contact) leaving the most correct portion intact.
- Occlusal contacts in maximum ICP should be composed of supporting cusp tips placed against flat or smoothly concave surfaces (or fossae) for stability.

  The use of articulating paper is continued until:
  - Heavy markings are no longer produced.
  - Contacts on the restoration have optimal position and form.
  - There is an even distribution of contacts on the casting and adjacent teeth.

Care must be exercised not to over-reduce the occlusal contacts.

Non-functional, non-working contacts on a restoration can be marked by inserting a strip of articulating paper and asking the patient to close in intercuspal position and then sliding the teeth laterally toward the opposite side. In a normal arrangement of teeth, contacts that might occur during a non-working pathway are positioned on the facial inclines

of the maxillary palatal cusps and the lingual inclines of the mandibular facial cusps. These non-working contacts must be removed with a suitable stone.

Now examine the casting for interferences in protrusive mandibular movements using the articulating paper. The areas that may have to be adjusted to prevent contact are the distal inclines of the maxillary cusps and mesial inclines of mandibular cusps.

Finally, interferences that occur on the casting between centric relation and intercuspal position are identified and corrected. Most patients have a small discrepancy between centric relation and maximum intercuspal position. Such a 'skid' is considered normal for most patients, but the operator should be sure that the casting does not have premature contact between centric relation and intercuspal position. Once the teeth have been marked in centric relation, observe the teeth to be sure that the casting does not have premature contacts in centric relation, and that it does not exacerbate any centric relation–intercuspal position skid. If it does, the mesial inclines of maxillary cusps and distal inclines of mandibular cusps will be the areas that need adjustment.

Various features of an optimum occlusion, which should be achieved in any occluding restoration are as follows:

1. There should be no tooth contact at the early stage of centric relation.
2. The mandible should arc along a hinge axis from centric relation to centric occlusion. Any interfering area within this arcing that create a translatory movement should be eliminated.
3. In the early stages of centric occlusion towards complete intercuspation, there may be forward movement of mandible. However, there should not be any lateral, medial or backward movement of mandible (with the teeth) when moving from centric relation to centric occlusion. Any cuspal incline contributing to this latter 'skid' should be eliminated.
4. At centric occlusion, the holding cusps should be of sufficient height to be in positive contact with their opposing counterparts. Such an arrangement preserves the vertical dimension of the teeth maintaining the stomatognathic system.
5. At centric occlusion, the holding markings should be symmetrical in magnitude and extent on all holding cuspal elements of both sides.
6. When maximum intercuspation is achieved, there should not be any further movement of mandible or the teeth.
7. Holding cusps occluding with more than one tooth or eccentrically occluding with opposing teeth should not move opposing teeth in a nonaxial direction. This is accomplished by broadening the contact of the cusps with opposing teeth to minimize tipping or wedging.
8. In centric occlusion, the incisal edges of lower incisors should be located at the gingival side of the lingual concavity of the upper incisors, preferably with a flat horizontal shelf or plane.
9. In lateral excursion of the mandible, there should not be any tooth contact on the non-working side of mandible.
10. In lateral excursion of mandible, the holding cusps of the working side should have a valley-like space on the opposing teeth (grooves or occlusal embrasures). This is to facilitate their non-interfering passage during lateral excursion.
11. The disclusion should start posteriorly and end by the cuspid's disclusion.
12. The disclusion path should be perfected so that the optimum direction along the lingual surface of the upper cuspid and the working inclines of the non-working cusps is achieved.
13. Marked contact areas during lateral excursion should be the same when going out of centric and back into centric.
14. During protrusive excursions of mandible, there should not be any tooth contact posteriorly.
15. Cuspids should be involved at least in the initial stages of the protrusive movement of mandible.
16. Protrusive contact markings should be evenly distributed and symmetrical on all teeth involved.

Occlusal equilibration of existing dentition should be performed before performing extensive restorative procedures; that will involve a change of occluding surfaces of multiple teeth. If symptoms of facial myodysfunction, temporomandibular joint dysfunction or dysfunction of any part of stomatognathic system are present: Such problems should also be dealt with before eliminating occlusal interferences.

The occlusion on the fabricated restorations is simulated in such a way that the restorations are in harmony with the stomatognathic system. Though this is quite a difficult procedure and also time consuming, yet the importance and advantages of establishing functional occlusion in a restoration warrant all clinicians to spend time and utilize expertise on occlusal principles.

The clinician should keep in mind that restorations, being affected by occlusal forces, should be managed accordingly; otherwise, may lead to treatment failures.

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